observed that the bacterial DNA in the heavy culture contained only $^{15}N$ and in the light culture only $^{14}N$. The heavy DNA could be distinguished from light DNA ($^{15}N$ from $^{14}N$) with a technique called **Cesium Chloride (CsCl) density gradient centrifugation**. In this process, heavy and light DNA extracted from cells in the two cultures settled into two distinct and separate bands (hybrid DNA) (**Fig. 5.5**).

The heavy culture ($^{15}N$) was then transferred into a medium that had only $NH_4Cl$, and took samples at various definite time intervals (20 minutes duration). After the first replication, they extracted DNA and subjected it to density gradient centrifugation. The DNA settled into a band that was intermediate in position between the previously determined heavy and light bands. After the second replication (40 minutes duration), they again extracted DNA samples, and this time found the DNA settling into two bands, one at the light band position and one at intermediate position. These results confirm Watson and Crick's semi conservative replication hypothesis.

## 5.8.2 Enzymes and mechanism of replication

In prokaryotes, replication process requires three types of DNA polymerases (DNA polymerase I, II, and III). DNA polymerase III is the main enzyme involved in DNA replication. DNA polymerase I (also known as **Kornberg enzyme**) and DNA polymerase II are involved in DNA repair mechanism. Eukaryotes have five types of DNA polymerases that catalyses the polymerization of nucleotides at the 3' OH of the new strand within a short period of time. *E. coli* that has only $4.6 \times 10^6$ bp, completes its replication process within 18 minutes; that means the average rate of polymerisation has to be approximately 2000 bp/sec. Replication takes place faster at the same time accurately. Any error will lead to mutation. However replication errors are corrected by repair enzymes such as nucleases. Deoxy nucleoside triphosphate acts as substrate and also provides energy for polymerization reaction.

Replication begins at the initiation site called the site of **'origin of replication' (ori)**. In prokaryotes, there is only one origin of replication, whereas in eukaryotes with giant DNA molecules, there can be several origins of replication (replicons). Since the two strands of DNA cannot be separated throughout at a time (due to large requirement of energy) the replication occurs within a small opening of the DNA helix called as replication fork. Unwinding of the DNA strand is carried out by DNA helicase. Thus, in one strand (template strand with polarity 3'⟶5') the replication is continuous and is known as the **leading strand** while in the other strand (coding strand with polarity 5'⟶3') replication is discontinuous, known as the **lagging strand (Fig. 5.6)**. The discontinuously synthesized fragments of the lagging strand (called the **Okazaki fragments**) are joined by the enzyme DNA ligase.

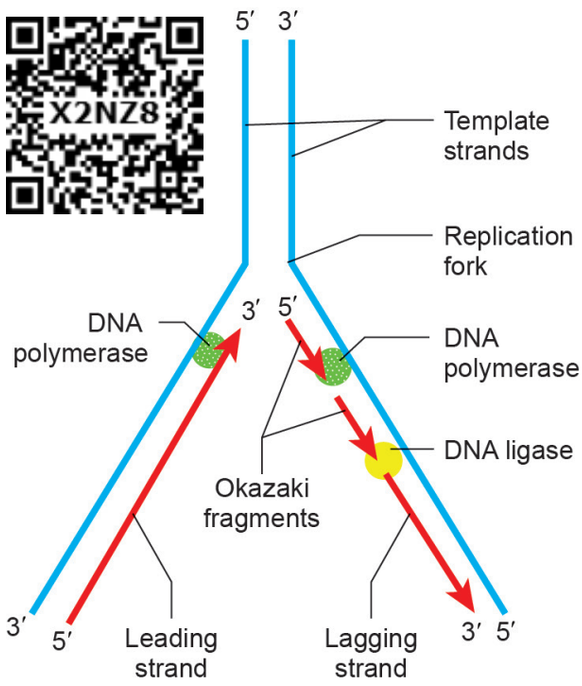



**Fig 5.6 Mechanism of replication showing a replication fork**



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